

ENONTEKIO KILPISJARVI, Finland

WMO: 028010

Lat: 69.049N

Lon: 20.791E

Elev: 483

StdP: 95.66

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-29.9	-35.7	0.1	-32.3	-32.8	0.2	-29.8	14.1	-4.7	12.7	-5.7	0.6	150	0.829

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	21.0	14.8	18.7	13.6	16.7	12.5	15.4	19.6	14.2	17.6	13.1	16.0	3.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	10.3	16.8	12.5	9.6	15.9	11.5	8.9	14.8	44.7	19.8	41.4	17.7	38.3	16.1	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.9	10.2	8.8	-35.6	23.6	2.1	2.1	-37.1	25.1	-38.3	26.3	-39.5	27.5	-41.0	29.1	
			DB												
			WB	-35.6	17.2	2.0	1.1	-37.0	18.0	-38.2	18.6	-39.4	19.3	-40.9	20.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-12.9	-12.6	-9.5	-3.6	2.8	7.6	11.9	10.4	6.1	-0.1	-6.4	-10.3
	DBStd	10.26	8.03	7.99	6.62	4.92	3.52	3.26	3.18	3.04	3.24	4.22	5.73	7.73
	HDD10.0	4267	709	632	605	408	225	86	16	32	121	314	492	628
	HDD18.3	7167	967	865	863	658	482	321	199	246	366	572	742	887
	CDD10.0	142	0	0	0	0	2	14	76	45	5	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	12	0	0	0	0	1	1	10	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.7	3.5	3.6	3.5	3.6	3.8	3.9	3.8	3.8	4.0	3.8	3.6	3.7
Precipitation	PrecAvg	477	52	35	30	21	26	41	69	48	34	42	34	41
	PrecMax	615	183	72	133	48	73	93	177	85	78	115	88	103
	PrecMin	280	13	6	7	3	6	2	9	24	7	16	4	1
	PrecStd	100	38	19	30	12	14	25	37	16	21	23	20	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	3.8	4.8	8.3	18.9	20.6	24.6	22.0	15.9	10.0	5.7	4.7
		MCWB	2.1	2.1	1.9	5.0	10.8	13.0	16.3	15.4	11.4	7.7	3.9	2.9
	2%	DB	1.6	2.0	2.4	5.9	13.6	17.5	22.2	19.7	13.9	8.6	4.0	2.7
		MCWB	0.4	0.3	0.8	3.0	8.0	11.5	15.4	14.8	10.6	6.9	2.8	1.3
	5%	DB	-0.3	0.3	0.8	4.5	10.1	15.5	19.9	17.3	12.3	7.2	1.9	0.6
		MCWB	-1.3	-0.8	-0.5	2.1	6.2	10.3	14.6	13.4	9.7	5.6	1.0	-0.5
	10%	DB	-2.3	-1.8	-0.6	3.2	8.0	13.6	17.6	15.4	10.9	5.8	0.5	-0.8
		MCWB	-3.1	-2.6	-1.8	1.3	4.9	9.6	13.1	12.3	8.7	4.4	-0.3	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.4	2.1	5.1	11.2	14.2	17.6	16.9	12.4	8.2	4.3	3.1
		MCDB	3.4	3.8	4.1	7.9	18.5	18.0	22.7	20.5	14.7	9.5	5.4	4.4
	2%	WB	0.6	0.4	0.9	3.3	8.1	12.6	16.0	15.0	11.1	7.0	2.8	1.4
		MCDB	1.5	1.7	2.3	5.4	12.6	16.1	21.1	18.4	13.3	8.3	4.0	2.6
	5%	WB	-1.3	-0.8	-0.4	2.3	6.4	11.2	14.9	13.8	10.1	5.8	1.1	-0.4
		MCDB	-0.3	0.3	0.8	4.2	9.8	14.8	18.8	16.8	11.9	7.2	1.8	0.5
	10%	WB	-3.0	-2.6	-1.8	1.3	5.1	9.9	13.8	12.6	9.0	4.4	-0.3	-1.7
		MCDB	-2.2	-1.8	-0.6	3.1	7.6	13.1	17.1	15.1	10.7	5.8	0.4	-0.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.3	9.4	8.9	6.7	7.2	7.5	6.9	5.6	4.4	6.0	7.9
		MCDBR	7.7	7.7	7.3	7.9	10.9	11.3	11.7	10.2	7.9	5.2	5.7	6.5
		MCWBR	7.2	7.1	6.3	5.8	6.5	6.3	6.2	5.9	5.3	4.2	5.0	6.0
	5% WB	MCDBR	7.7	7.5	7.2	7.0	10.5	10.0	10.1	9.3	7.0	5.1	5.6	6.7
		MCWBR	7.2	7.0	6.3	5.3	6.5	6.0	5.6	5.7	4.9	4.2	4.9	6.2
Clear-Sky Solar Irradiance	taub	0.183	0.235	0.281	0.304	0.321	0.325	0.330	0.317	0.288	0.245	0.183	N/A	
	taud	1.902	2.061	2.139	2.189	2.291	2.468	2.491	2.509	2.547	2.216	1.952	N/A	
	Ebn at Noon	47	526	733	833	864	874	856	829	765	517	39	0	
	Edh at Noon	5	43	79	102	105	91	87	76	55	36	4	0	
All-Sky Solar Radiation	RadAvg	0.02	0.39	1.77	3.75	5.00	4.93	4.56	3.24	1.73	0.53	0.07	0.00	
	RadStd	0.00	0.04	0.10	0.20	0.40	0.40	0.40	0.25	0.16	0.06	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page